

Forecasting Student Enrollment Using Simple Moving Average (SMA), Weighted Moving Average (WMA), and Exponential Moving Average (EMA)

**with Error Analysis Using Mean Squared Error
(MSE)**

Lab Report / Experiment

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1 Title of Experiment

Forecasting Student Enrollment Using Simple Moving Average (SMA), Weighted Moving Average (WMA), and Exponential Moving Average (EMA) with Error Analysis Using Mean Squared Error (MSE).

2 Aim / Objectives

The aim of this experiment is to apply and compare three time-series forecasting techniques Simple Moving Average, Weighted Moving Average, and Exponential Moving Average - on historical student enrollment data, and to evaluate their accuracy using Mean Squared Error.

Specific Objectives

- To calculate the 3-year Simple Moving Average (SMA) for student enrollment data from 2018 to 2025.
- To calculate the 3-year Weighted Moving Average (WMA) using weights of 1, 2, and 3.
- To calculate the Exponential Moving Average (EMA) using a smoothing factor $\alpha = 0.4$.
- To compute the Mean Squared Error (MSE) for each forecasting method and compare accuracy.
- To plot all three forecasting curves alongside actual enrollment on a single graph in Microsoft Excel.
- To forecast the enrollment value for the year 2026 using each method.

3 Theory / Introduction

Forecasting is the process of predicting future values of a variable based on its past observations. In operations research and management science, accurate forecasting is essential for planning resources, budgets, and capacity. Time series forecasting methods analyze historical data collected over regular time intervals to identify patterns and predict future values.

This experiment uses student enrollment records from 2018 to 2025 and applies three widely used smoothing techniques.

3.1 Simple Moving Average (SMA)

Simple Moving Average (SMA) is the average of observed data values over a specific number of periods. It smooths fluctuations in data and helps identify trends.

$$\text{SMA} = \frac{A_1 + A_2 + \cdots + A_n}{n}$$

Where:

- A = Observed values
- n = Number of periods

3.2 Weighted Moving Average (WMA)

Weighted Moving Average (WMA) assigns higher importance to recent observations by applying weights to the data points.

$$\text{WMA} = \frac{(P_1 \times W_1) + (P_2 \times W_2) + \cdots + (P_n \times W_n)}{W_1 + W_2 + \cdots + W_n}$$

Where:

- P = Data point
- W = Weight assigned to each data point
- n = Number of periods

3.3 Exponential Moving Average (EMA)

Exponential Moving Average (EMA) gives more importance to recent observations using a smoothing factor.

$$\text{EMA}_t = \alpha Y_t + (1 - \alpha) \text{EMA}_{t-1}$$

Where:

- α = Smoothing factor
- Y_t = Current observation
- EMA_{t-1} = Previous EMA value

3.4 Mean Squared Error (MSE)

Mean Squared Error (MSE) measures the average squared difference between actual and predicted values.

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Where:

- y_i = Actual value
- $\hat{y}_i$ = Forecasted value
- n = Number of observations

4 Observation and Calculations

The student enrollment data from 2018 to 2025 was analyzed using SMA, WMA, and EMA forecasting techniques in Microsoft Excel. Error values and Mean Squared Error (MSE) were calculated for each forecasting method.

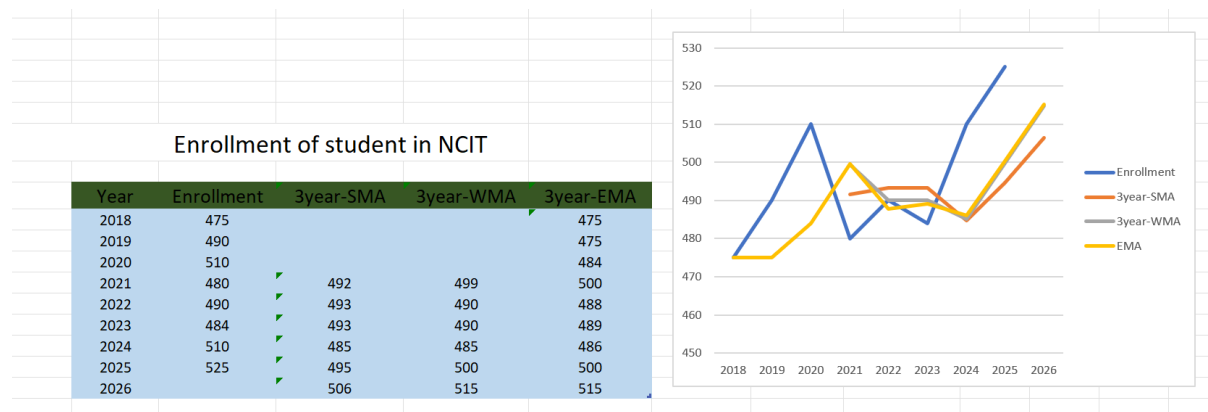


Figure 1: Forecasting Calculations and Graph for SMA, WMA, and EMA

MSE for SMA			
Enrollment	3year-SMA	Error	Error^2
475			
490			
510			
480	492	-12	144
490	493	-3	9
484	493	-9	81
510	485	25	625
525	495	30	900
	506		
MSE for SMA =			351.8

Figure 2: MSE Calculation for SMA

MSE for WMA			
Enrollment	3year-WMA	Error	Error^2
475			
490			
510			
480	499	-19	361
490	490	0	0
484	490	-6	36
510	485	25	625
525	500	25	625
	515		
MSE for WMA =			329.4

Figure 3: MSE Calculation for WMA

MSE for EMA			
Enrollment	3year EMA	Error	Error^2
475	475	0	0
490	475	15	225
510	484	26	676
480	500	-20	400
490	488	2	4
484	489	-5	25
510	486	24	576
525	500	25	625
	515		
MSE of EMA =			316.375

Figure 4: MSE Calculation for EMA

5 Results

The forecasted enrollment values for the year 2026 were obtained using different forecasting techniques as follows:

- Forecast using SMA = 506
- Forecast using WMA = 515
- Forecast using EMA = 515

The Mean Squared Error values obtained were:

- MSE for SMA = 351.8
- MSE for WMA = 329.4
- MSE for EMA = 316.375

Among all forecasting methods, EMA produced the lowest MSE value and therefore provided the most accurate forecast.

6 Conclusion

In this experiment, three forecasting techniques — Simple Moving Average (SMA), Weighted Moving Average (WMA), and Exponential Moving Average (EMA) — were applied to student enrollment data.

The forecasting values and error analysis showed that the Exponential Moving Average (EMA) method produced the lowest Mean Squared Error (MSE), indicating higher forecasting accuracy compared to SMA and WMA.

Thus, EMA is considered the most effective forecasting technique for this dataset because it gives more importance to recent observations and responds more effectively to changes in enrollment trends.